

GRADE 10 ICT

Lesson 5

Operating Systems

MODULE 5.1

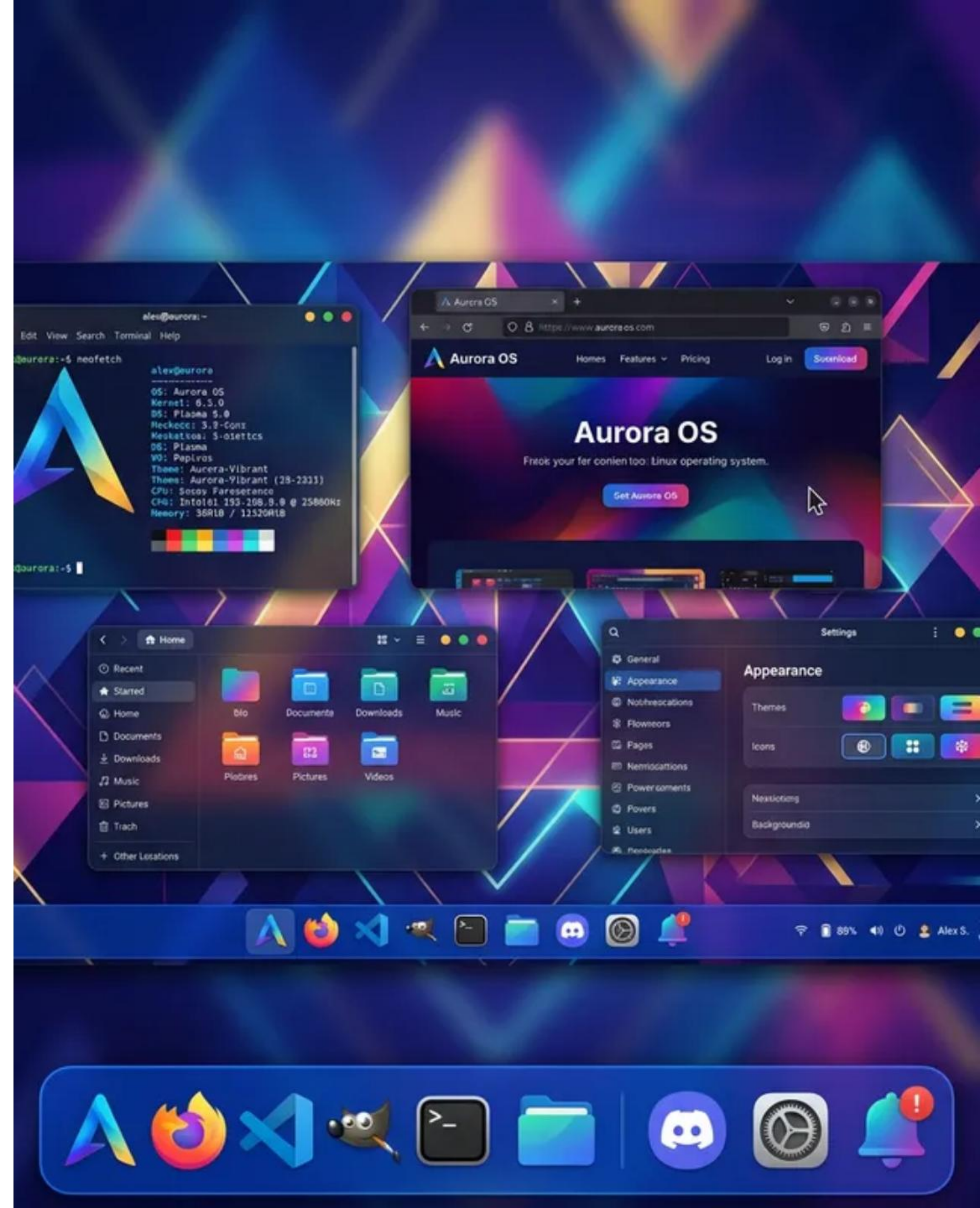
Introduction to Operating Systems & Functions of an Operating System

What is an Operating System?

An Operating System (OS) is a set of programs that acts as a link between the user and the computer hardware.

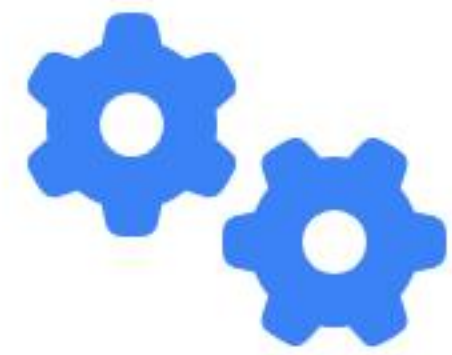
User ↔ Operating System ↔ Hardware

Without an OS, a computer cannot function.



Software Types

Computer software is divided into two main categories.



System Software

Controls computer operations and manages hardware.

Example: Operating System



Application Software

Programs designed to do specific tasks for the user.

Examples: Word, Games

Examples of Operating Systems



Windows



macOS / iOS



Linux (Ubuntu)



Android

Types of Operating Systems

OS Type	Description
Single User	Allows only one user to use the computer at a time (e.g., DOS).
Multi User	Allows multiple users to access the computer at the same time (e.g., Linux, Windows Server).
Multi-tasking	Allows multiple programs to run at the same time (e.g., Windows 10, macOS).
Real Time	Responds to input instantly without delay (e.g., ATMs, traffic control systems).

Functions of an Operating System

The OS provides essential services to keep the computer running smoothly.



Process Management



Memory Management



Device Management



File Management



Security Management



Network Management

Core Functions Explained

Process Management

The OS decides which program gets to use the CPU and for how long.

- ✓ **Allocates processor time**
- ✓ **Keeps tasks running smoothly**

Memory Management

The OS manages the computer's primary memory (RAM).

- ✓ **Gives memory space to running apps**
- ✓ **Frees memory when apps close**

Managing Hardware & Data

Device Management

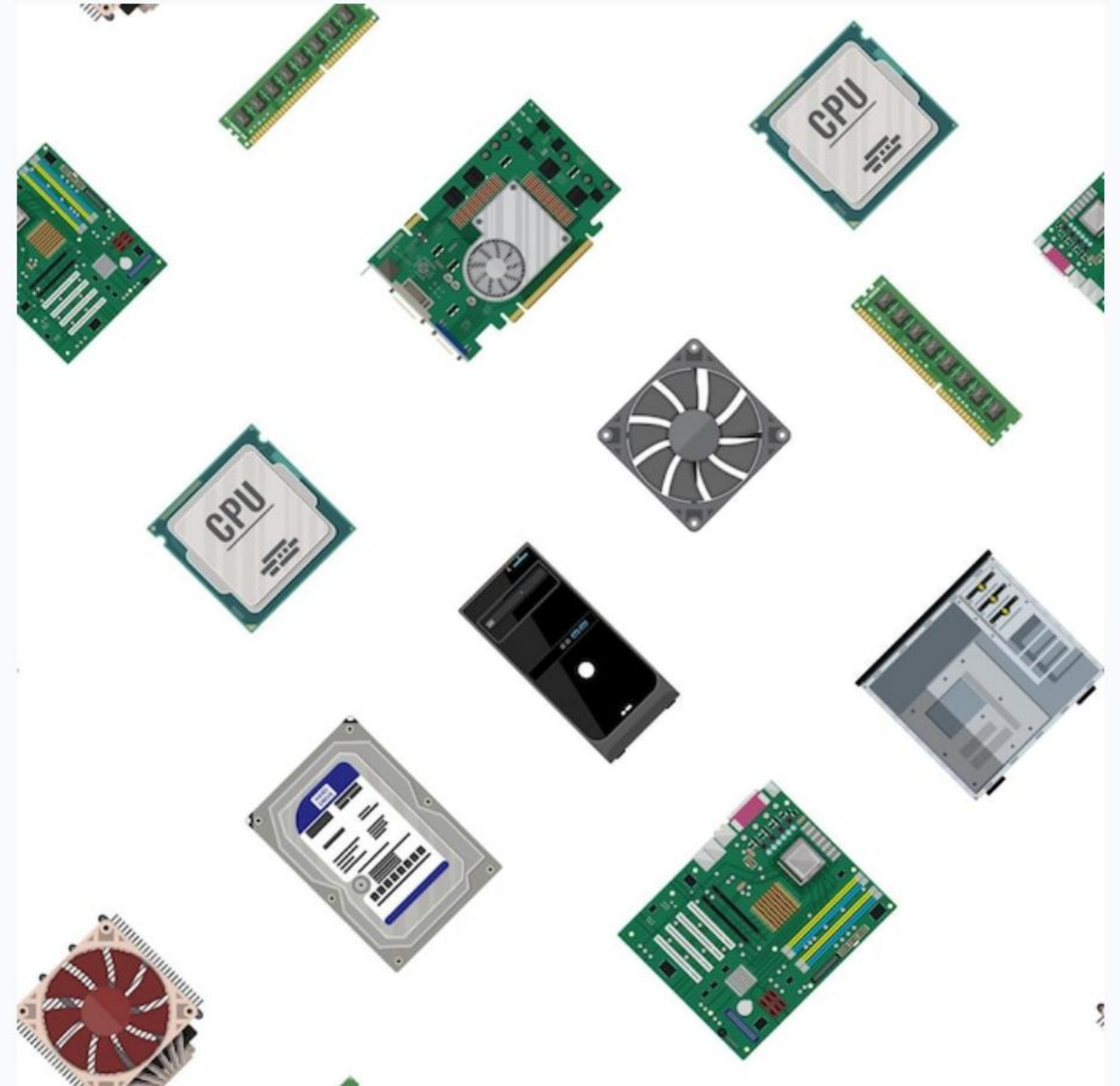
Controls input and output devices (printers, keyboards) using special software called drivers.

File Management

Organizes how files and folders are saved, found, and deleted on storage drives.

Security Management

Protects data using passwords and user accounts.



Module 5.1 Quick Review

Q: What links the user to the hardware?

A: The Operating System (OS).

Q: Give two examples of OS.

A: Windows, Android, macOS.

Q: Managing RAM is part of which function?

A: Memory Management.